

At the same time, the Vigenère cipher doesn't suffer from the frequency analysis weakness that can crack the simple substitution cipher. For centuries, the Vigenère cipher represented the state of the art in cryptography.

You'll notice many similarities between the code for the Vigenère and Caesar cipher programs. More info about the Vigenère cipher can be found at https://en.wikipedia.org/wiki/Vigen%C3%A8re_cipher. If you'd like to learn more about ciphers and code breaking, you can read my book *Cracking Codes with Python* (No Starch Press, 2018; <https://nostarch.com/crackingcodes/>).

The Program in Action

When you run *vigenere.py*, the output will look like this:

```
Vigenère Cipher, by Al Sweigart al@inventwithpython.com
The Vigenère cipher is a polyalphabetic substitution cipher that was
powerful enough to remain unbroken for centuries.
Do you want to (e)ncrypt or (d)ecrypt?
> e
Please specify the key to use.
It can be a word or any combination of letters:
> PIZZA
Enter the message to encrypt.
> Meet me by the rose bushes tonight.
Encrypted message:
Bmds mt jx sht znre qcrgeh bnmivps.
Full encrypted text copied to clipboard.
```

How It Works

Because the encryption and decryption processes are fairly similar, the `translateMessage()` function handles both of them. The `encryptMessage()` and `decryptMessage()` functions are merely *wrapper functions* for `translateMessage()`. In other words, they are functions that adjust their arguments, forward these to another function, and then return that function's return value. This program uses these wrapper functions so that they can be called in a manner similar to `encryptMessage()` and `decryptMessage()` in Project 66, "Simple Substitution Cipher." You can import these projects as modules in other programs to make use of their encryption code without having to copy and paste the code directly into your new program.

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1. `"""Vigenère Cipher, by Al Sweigart al@inventwithpython.com`
 2. `The Vigenère cipher is a polyalphabetic substitution cipher that was`
 3. `powerful enough to remain unbroken for centuries.`
 4. `More info at: https://en.wikipedia.org/wiki/Vigen%C3%A8re\_cipher`
 5. `View this code at https://nostarch.com/big-book-small-python-projects`
 6. `Tags: short, cryptography, math"""`
 7.
 8. `try:`
 9. `import pyperclip # pyperclip copies text to the clipboard.`